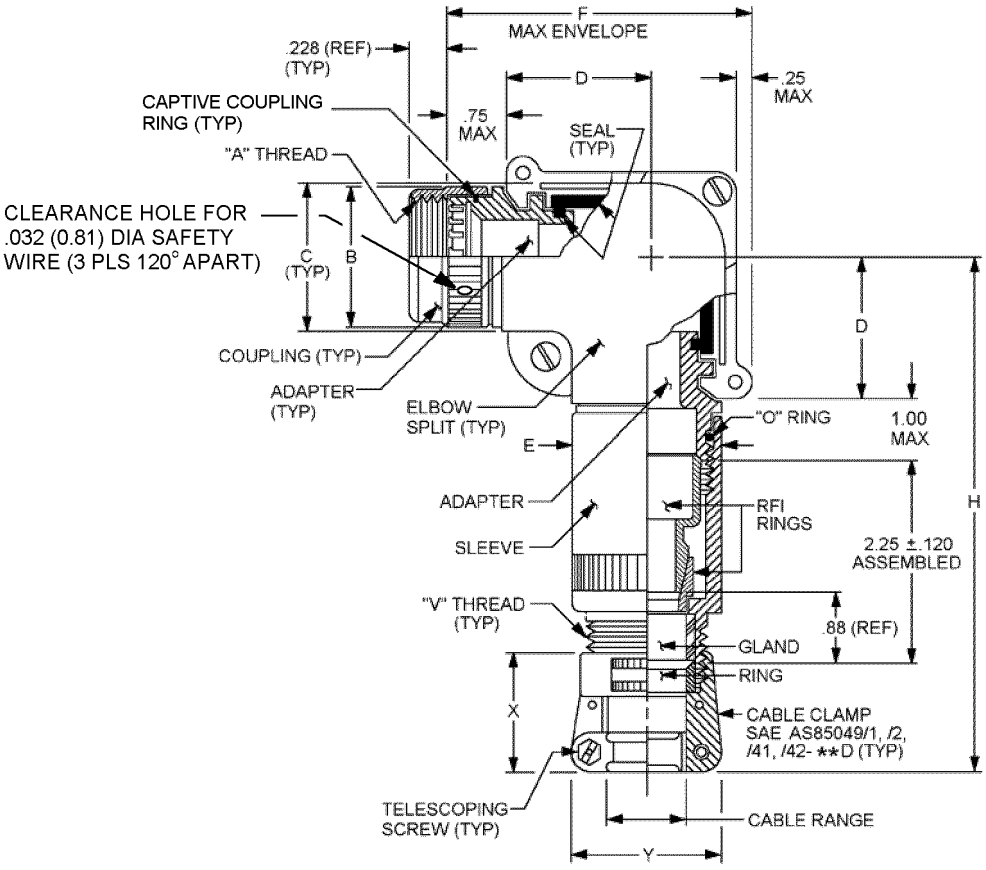
INCH-POUND
MIL-DTL-28840/8D
26 April 2007
SUPERSEDING
MIL-DTL-28840/8C
20 April 2004

DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, CIRCULAR, THREADED, HIGH SHOCK, HIGH DENSITY, SHIPBOARD, BACKSHELL, 90°, CABLE SEALING AND SHIELD TERMINATION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-28840.



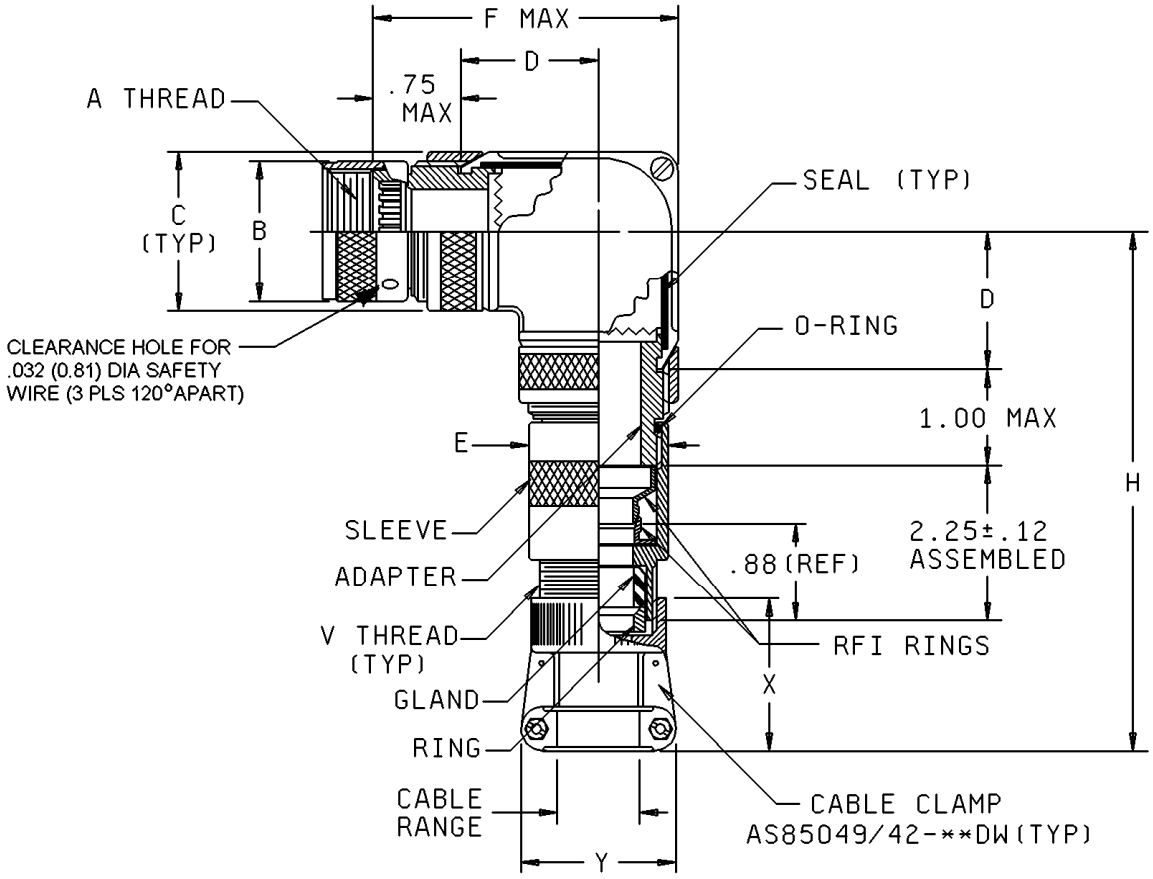
Style A

FIGURE 1. Dimensions and configurations.

AMSC N/A

FSC 5935

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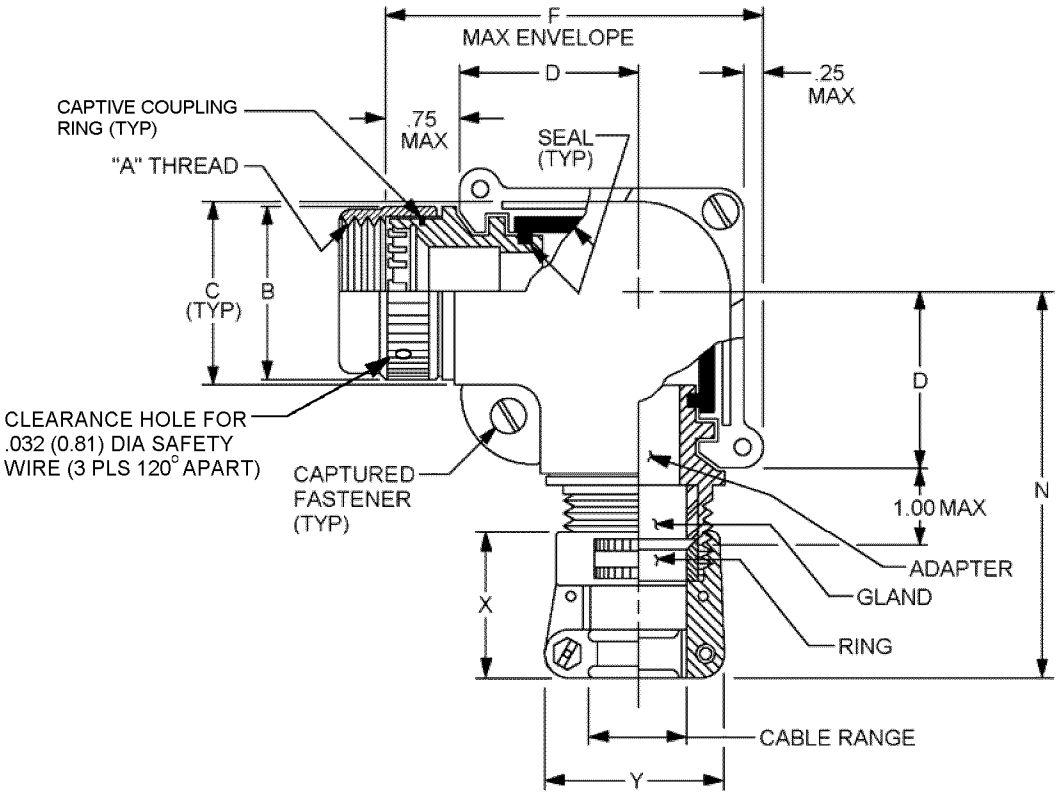


ALTERNATE STYLE A

Inches	mm
.12	3.05
.228	5.79
.25	6.35
.75	19.05
.88	22.35
2.25	57.15

FIGURE 1. Dimensions and configurations - Continued.

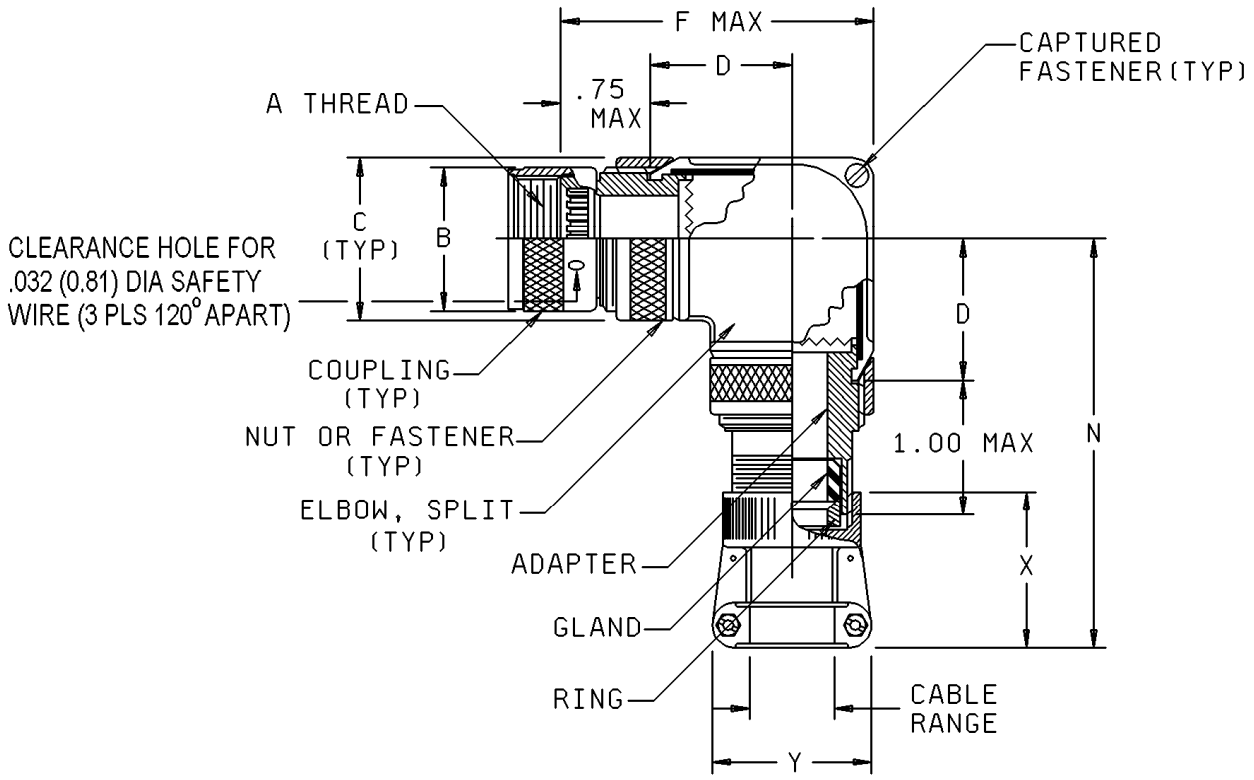
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Style B

FIGURE 1. Dimensions and configurations - Continued.

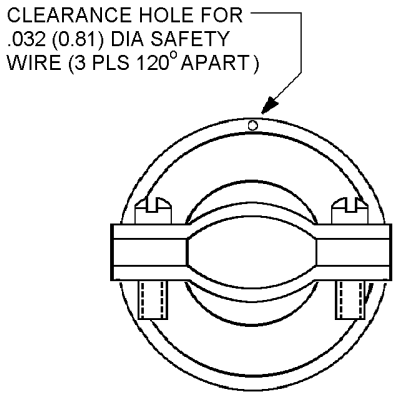
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ALTERNATE STYLE B

FIGURE 1. Dimensions and configurations - Continued.

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Backshell size (see note 3)	Dash no.	Designator (see note 5)	A thread class 2B	B max dia	C max dia	D max dia
A	01	A (11)	.750-20 UNEF	1.028 (26.11)	1.188 (30.17)	1.031 (26.19
A	02	B (13)	.875-20 UNEF	1.141 (28.98)		1.031 (26.19)
B	03	B (13)				
A	04	C (15)	1.00-20 UNEF	1.263 (32.08)	1.438 (36.52)	1.156 (29.36)
B	05	C (15)				
C	06	C (15)				
D	07	C (15)			1.688 (42.87)	1.281 (32.54)
A	08	D (17)	1.125-18 UNEF	1.387 (35.23)		
B	09	D (17)				
C	10	D (17)				
A	11	E (19)	1.250-18UNEF	1.513 (38.43)		
B	12	E (19)			1.938 (49.22)	1.406 (35.71)
A	13	F (23)	1.438-18 UNEF	1.703 (43.51)		
B	14	F (23)				
C	15	F (23)				
A	16	G (25)	1.562-18 UNEF	1.825 (46.35)		
B	17	G (25)				
C	18	G (25)			2.188 (54.43)	1.531 (38.89)
D	19	G (25)				
A	20	H (29)	1.875-16 UN	2.143 (54.43)		
B	21	H (29)				

See notes at end of figure.

FIGURE 1. Dimensions and configurations - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:34Z
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Back shell size (see note 3)	Dash no.	Designator (see note 5)	Cable range		E max dim	F max dim	V thread class 2/
			Max	Min			
A	01	A (11)	.312 (7.92)	.188 (4.77)	1.156 (29.36)	2.614 (66.40)	.625-24 UNEF
A	02	B (13)	.312 (7.92)	.188 (4.77)	1.156 (29.36)		.625-24 UNEF
B	03	B (13)	.438 (11.12)	.281 (7.14)			.750-20 UNEF
A	04	C (15)	.312 (7.92)	.188 (4.77)	1.281 (32.54)	2.864 (72.75)	.625-24 UNEF
B	05	C (15)	.438(11.12)	.281 (7.14)			.750-20 UNEF
C	06	C (15)	.562 (14.27)	.344 (8.74)			.875-20 UNEF
D	07	C (15)	.750 (19.05)	.438 (11.12)	1.406 (35.71)	3.114 (79.10)	1.188-18 UNEF
A	08	D (17)	.625 (15.87)	.375 (9.82)			1.000-20 UNEF
B	09	D (17)	.750 (19.05)	.438 (11.12)			1.188-18 UNEF
C	10	D (17)	.938 (23.82)	.625 (15.87)	1.531 (38.89)		1.438-18 UNEF
A	11	E (19)	.625 (15.87)	.375 (9.52)	1.406 (35.71)		1.000-20 UNEF
B	12	E (19)	.938 (23.82)	.625 (15.87)	1.531 (38.89)	3.364 (85.45)	1.438-18 UNEF
A	13	F (23)	.750 (19.05)	.438 (11.12)	1.406 (25.71)		1.188-18 UNEF
B	14	F (23)	.938 (23.82)	.625 (15.87)	1.531 (38.89)		1.438-18 UNEF
C	15	F (23)	1.188 (30.17)	.812(20.62)	1.781 (45.24)		1.750-18 UNS
A	16	G (25)	.938 (23.82)	.625 (15.87)	1.531 (38.89)		1.438-18 UNEF
B	17	G (25)	1.188 (30.17)	.812 (20.62)	1.781 (45.24)		1.750-18 UNS
C	18	G (25)	1.250 (31.75)	.875 (22.23)		3.614 (91.80)	1.750-18 UNS
D	19	G (25)	1.380 (35.05)	1.000 (25.40)	2.156 (54.76)		2.000-16 UNS
A	20	H (29)	1.188 (30.17)	.812 (20.62)	1.781 (45.24)		1.750-18 UNS
B	21	H (29)	1.380 (35.05)	1.000 (25.40)	2.156 (54.76)		2.000-18 UNS

See notes at end of figure.

FIGURE 1. Dimensions and configurations - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:34Z
Check the source to verify that this is the current version before use.

Back shell size (see note 3)	Dash no.	Designator (see note 5)	X ref dim	Y ref dim	H max style A	N max style B
A	01	A (11)	1.281 (32.54)	.938 (23.82)	5.332 (134.43)	2.962 (75.23)
A	02	B (13)	1.281 (32.54)	.938 (23.82)	5.332 (134.43)	2.962 (75.23)
B	03	B (13)	1.281 (32.54)	1.125 (28.57)	5.332 (134.43)	2.962 (75.23)
A	04	C (15)	1.281 (32.54)	.938 (23.82)	5.457 (138.61)	3.087 (78.41)
B	05	C (15)	1.281 (32.54)	1.125 (28.57)	5.457 (138.61)	3.087 (78.41)
C	06	C (15)	1.281 (32.54)	1.312 (33.32)	5.457 (138.61)	3.087 (78.41)
D	07	C (15)	1.281 (32.54)	1.593 (40.62)	5.682 (144.32)	3.312 (84.12)
A	08	D (17)	1.281 (32.54)	1.312 (33.32)	5.682 (144.32)	3.312 (84.12)
B	09	D (17)	1.312 (33.32)	1.593 (40.62)	5.713 (145.11)	3.343 (84.91)
C	10	D (17)	1.406 (35.71)	1.750 (44.45)	5.807 (147.50)	3.437 (87.30)
A	11	E (19)	1.281 (32.54)	1.312 (33.32)	5.682 (144.32)	3.312 (84.12)
B	12	E (19)	1.406 (35.71)	1.750 (44.45)	5.832 (148.13)	3.462 (87.93)
A	13	F (23)	1.312 (33.32)	1.593 (40.62)	5.738 (145.25)	3.368 (85.55)
B	14	F (23)	1.406 (35.71)	1.750 (44.45)	5.832 (148.13)	3.462 (87.93)
C	15	F (23)	1.593 (40.46)	2.093 (53.16)	6.019 (152.88)	3.649 (92.68)
A	16	G (25)	1.406 (35.71)	1.750 (44.45)	5.832 (148.13)	3.462 (87.93)
B	17	G (25)	1.593 (40.46)	2.093 (53.16)	6.019 (152.88)	3.649 (92.68)
C	18	G (25)	1.593 (40.46)	2.093 (53.16)	6.144 (156.06)	3.774 (95.86)
D	19	G (25)	1.625 (41.28)	2.343 (59.51)	6.176 (156.87)	3.806 (96.62)
A	20	H (29)	1.593 (40.46)	2.093 (53.16)	6.144 (156.06)	3.774 (95.86)
B	21	H (29)	1.625 (41.28)	2.343 (59.51)	6.176 (156.87)	3.806 (96.62)

See notes at end of figure.

FIGURE 1. Dimensions and configurations - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:34Z
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Back shell size (see note 3)	Dash no.	Designator (see note 5)	A thread class 2B	B max dia	C max dia	D max dia
A	22	J (33)	2.062-16 UNS	2.329 (59.16)	2.688 (68.26)	1.781 (45.24)
A	23	J (33)				
B	24	A (11)	.750-20 UNEF	1.028 (26.11)	1.188 (30.17)	1.031 (26.19)
C	25	A (11)			1.438 (36.52)	1.156 (29.36)
C	26	B (13)	.875-20 UNEF	1.141 (28.98)		
D	27	B (13)			1.688 (42.87)	1.281 (32.54)
E	28	C (15)	1.000-20 UNEF	1.263 (32.08)		

Back shell size (see note 3)	Dash no.	Designator (see note 5)	Cable range		E max dim	F max dim	V thread class 2/
			max	min			
A	22	J (32)	1.380 (35.05)	1.000 (25.40)	2.281 (57.94)	4.114 (104.50)	2.000-18 UNS
A	23	J (33)	1.625 (41.26)	1.250 (31.75)			2.250-16 UNS
B	24	A (11)	438 (11.12)	281 (7.14)	1.156 (29.36)	2.614 (66.40)	.750-20 UNEF
C	25	A (11)	562 (14.27)	344 (8.74)	1.281 (32.54)	2.864 (72.75)	.875-20 UEF
C	26	B (13)	562 (14.27)	344 (8.74)	1.281 (32.54)		.875-20 UEF
C	27	B (13)	625 (15.87)	375 (9.52)	1.406 (35.71)	3.114 (79.10)	1.000-20 UNEF
B	28	C (15)	625 (15.87)	375 (9.50)			1.000-20 UNEF

See notes at end of figure.

FIGURE 1. Dimensions and configurations - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:34Z
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Back shell size (see note 3)	Dash no.	Designator (see note 5)	X ref dim	Y ref dim	H max style A	N max style B
A	22	J (33)	1.625 (41.28)	2.343 (59.51)	6.426 (163.22)	4.056 (103.02)
A	23	J (33)	1.900 (48.26)	2.750 (69.85)	6.701 (170.21)	4.331 (110.01)
B	24	A (11)	1.281 (32.54)	1.125 (28.57)	5.332 (135.43)	2.962 (75.23)
C	25	A (11)	1.281 (32.54)	1.312 (33.32)	5.457 (138.61)	3.087 (78.41)
C	26	B (13)	1.281 (32.54)	1.312 (33.32)	5.457 (138.61)	3.087 (78.41)
C	27	B (13)	1.281 (32.54)	1.312 (33.32)	5.682 (144.32)	3.312 (84.12)
B	28	C (15)	1.281 (32.54)	1.312 (33.32)	5.682 (144.32)	3.312 (84.12)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Designator of backshell size within each shell size.
4. Dimensions apply after plating.
5. Shell sizes are provided within parentheses for information and are not a part of the designator.

FIGURE 1. Dimensions and configurations - Continued.

REQUIREMENTS

Dimensions and configurations: See figure 1.
Interface dimensions shall conform to MIL-DTL-28840.

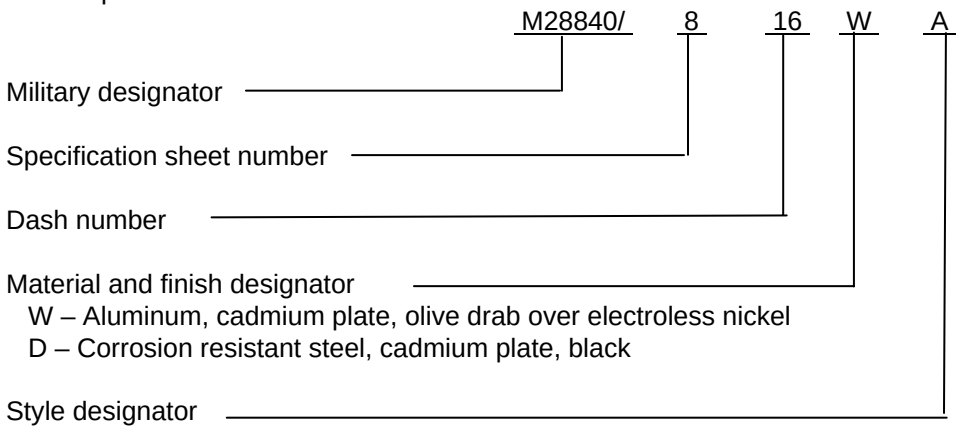
Material: Shall be aluminum alloy in accordance with SAE-AMS-QQ-A-225, ASTM B85, ASTM B108, or ASTM B26/B26M. Corrosion resistant steel shall be in accordance with ASTM A276, ASTM A580/A580M or ASTM A473, type 316.

Finish: Aluminum components shall be cadmium plate, olive drab over electroless nickel.
Corrosion resistant steel components shall be cadmium plate, black. The screws and washers shall be passivated in accordance with SAE-AMS-2700, type 2.

Marking: The backshell shall be identified with the manufacturer's name or trademark and the Part or Identifying Number (PIN).

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PIN example:



Qualification: Two samples in each shell size for which qualification is desired and shall be tested in accordance with MIL-DTL-28840.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Reference documents. In addition to MIL-DTL-28840, this document references the following:

ASTM A276	ASTM B85	SAE AS85049/1
ASTM A473	ASTM B108	SAE AS85049/2
ASTM A580/A580M	SAE-AMS-QQ-A-225	SAE AS85049/41
ASTM B26/B26M	SAE-AMS2700	SAE AS85049/42

CONCLUDING MATERIAL

Custodians:
Navy - EC
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-2005-032)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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